Cloud-Guided Fusion With SAR-to-Optical Translation for Thick Cloud Removal

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***Abstract*— Deep learning has been widely used in thick cloud removal (TCR) for optical satellite images. Since thick clouds completely block the surface, synthetic aperture radar (SAR) images have recently been used to assist in the recovery of occluded information. However, this approach faces several challenges: 1) the significant domain gap between SAR and optical features can cause interference in recovering occluded optical information from SAR and 2) the TCR methods need to distinguish between cloudy and non-cloudy regions; otherwise, inconsistencies may arise between the recovered regions and the remaining non-cloudy regions. To this end, we propose a new SAR-assisted TCR method based on a two-step fusion framework, which consists of the feature alignment translation (FAT) network and the cloud-guided fusion (CGF) network. First, the FAT leverages the common features between SAR and optical images to translate SAR images into corresponding optical images, thus recovering the occluded information. Considering the gap between the translated images and the real cloud-free images, the CGF utilizes cloudy images to further refine the translated images, resulting in the cloud-removed images. In the CGF, cloud distribution is predicted to distinguish between cloudy and non-cloudy regions. Then, the cloud distribution is used to guide the refinement of recovered regions using non-cloudy regions. Extensive experiments on both simulated and real datasets show that the proposed algorithm achieves better performance compared with the state-of-the-art methods.**

***Index Terms*— Cloud removal, data fusion, deep learning, generative adversarial network (GAN), synthetic aperture radar (SAR)-to-optical translation.**

I. Introduction

OPTICAL satellite images are essential in Earth observation but are easily obstructed by clouds. According to a study, clouds obstruct an average of 55% land surface during a seasonal cycle, limiting subsequent analysis [21]. Therefore, cloud removal is necessary to restore the covered information. Clouds include thin and thick clouds. The satellite sensors can still capture some reflected light when thin clouds cover the surface. However, thick clouds can completely block the surface, leading to missing information. As a result, thick cloud removal (TCR) is more challenging and has received considerable attention in recent cloud removal studies.

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In TCR, auxiliary data is used to compensate for the absence of information under thick clouds. According to the type of the auxiliary data, existing TCR methods consist of multitemporal methods [3], [4], [5], [6], [7], [8], [9], [10] and multisource methods [1], [2], [11], [12], [13], [14], [15], [16], [17], [18], [20]. Multitemporal methods use images taken from the same sensor at different times as auxiliary data. These methods perform well under short time intervals when there are no discernible changes in land cover. Nonetheless, satellites like Landsat and Sentinel-2, characterized by long temporal resolutions, do not always ensure short time intervals between captures [11]. On the other hand, multisource methods utilize images from different sensors as auxiliary data. Recently, synthetic aperture radar (SAR) has emerged as a popular auxiliary source due to its superior penetration capability, allowing SAR images to capture information hidden by clouds [2]. Furthermore, if a satellite platform is equipped with both SAR and optical sensors, it becomes feasible to acquire SAR images concurrently with optical imagery. Therefore, SAR-assisted multisource methods possess great potential for TCR.

In recent years, deep learning has been effectively employed in TCR and improved the performance. In TCR, a prevalent deep learning approach is the one-step fusion method [12], [16], [17], [20], [26], where both cloudy and SAR images are inputted into a fusion network to generate cloud-removed images [as shown in Fig. 1(a)]. Similar to the common fusion, the one-step fusion aims to provide the results using the advantages of different modalities simultaneously. Specifically, the one-step fusion expects the cloud-removed images to fuse the characteristics of SAR images that are not affected by clouds and the spectral information of optical images. In common fusion tasks, such as infrared and visible image fusion, there is spatial correspondence between the input images, and the output modality is different from the input modality. However, TCR is different from the common fusion tasks. On one hand, the SAR and optical images exhibit correspondence in non-cloudy regions, but not in cloudy regions. On the other hand, the cloud-removed images maintain the same modality as the input optical images. As a result, the one-step fusion network may need to learn both cloud segmentation and SAR-to-optical translation internally. Since these two tasks are quite different, it is challenging to master all tasks effectively in one network, potentially resulting in distortions in the restored information.

The two-step fusion framework [1], [2] has recently exhibited its superiority in SAR-assisted TCR. It first translates the SAR images into optical images by a translation network. Subsequently, it fuses the translated images with cloudy

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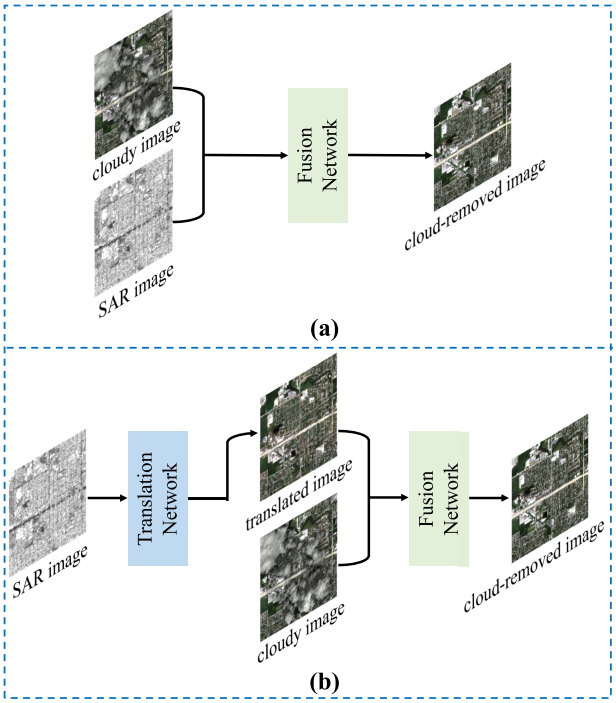


Fig. 1. Two types of deep learning framework for SAR-assisted TCR. (a) One-step fusion. (b) Two-step fusion.

images using a fusion network to produce cloud-removed images [as shown in Fig. 1(b)]. In contrast to the one-step fusion, the two-step fusion learns the SAR-to-optical translation and image fusion separately, thus mitigating the learning complexity. Despite the advantages, there are two problems in the existing two-step fusion methods.

1) For the translation network in the first step, the translated images exhibit structure and color distortions. One possible reason is that the common features in network are interfered and weakened. Common features are those effective features that are shared across different modalities (i.e., SAR and optical modalities). The common features not only assist the network in learning the mapping from SAR to optical spectra but also influence the quality of the translated images. For example, structural features may be one of the common features between SAR and optical images. If the structural features undergo interference during the translation, the resulting translated images can suffer from structural distortions.
2) For the fusion network in the second step, the cloud-removed images lack details in the recovered regions. One possible reason is that the current methods ignore the distinctions between cloudy and non-cloudy regions. First, the information corresponding to these regions in the translated images is not equally significant; there may be redundant data in the non-cloudy regions. Existing methods concatenate the translated and cloudy images along channels before feeding them into subsequent networks, potentially introducing excessive redundancy. Second, features in cloudy regions should be optimized, while features in non-cloudy regions should be retained and used for guiding the optimization. However, the commonly used convolution operation treats all positions equally, resulting in limited restoration performance.

To overcome the problems, a new SAR-assisted TCR method is proposed based on the two-step fusion framework, which consists of a feature alignment translation (FAT) network and a cloud-guided fusion (CGF) network. In the FAT, a feature pyramid encoder (FPE) is first designed to extract multiscale features of SAR images. To mitigate the interference of SAR speckle noise on low-scale features, the FPE adopts a strategy of transmitting deep semantic features to the shallower low-scale features, thereby increasing the proportion of effective features among low-scale features. The FPE ensures that the extracted features can contain sufficient common features. Subsequently, the multiscale SAR features are fused and gradually mapped into the translated images. However, given the disparities between features at different scales, the distribution of common features may change after the feature fusion. To this end, an alignment fusion module (AFM) is proposed to align the distribution of common features before and after the fusion. In the AFM, the distribution of common features is first predicted, and then this distribution is utilized to strengthen the corresponding positions in fused features, thus aligning the distributions. Considering that the translated images from the FAT still exhibits a gap with the cloud-free images, the CGF utilizes cloudy images to further refine the restored information in the translated images.

In the CGF, in order to reduce the redundant information from non-cloudy regions in the translated images, a cloud-aware fusion module (CFM) is designed. This module first predicts a cloud distribution map to differentiate between cloudy and non-cloudy regions, subsequently attenuating the non-cloudy regions in the translated images while retaining the restoration information in cloudy regions. To treat the features in cloudy and non-cloudy regions differently, a region gated module (RGM) is designed. With the cloud distribution map, the RGM can dynamically process features between cloudy and non-cloudy regions through a gating mechanism. The overview of the proposed method is illustrated in Fig. 2.

We conducted experiments using the real-world SEN12MS-CR dataset [17]. Furthermore, a simulated dataset, SIM-CR, was constructed to further validate the effectiveness of the proposed method. The experimental results demonstrate that the proposed method outperforms the existing methods in TCR.

In summary, the contributions of this work are as follows.

1) We propose a new SAR-assisted TCR method, which consists of a new translation network and a new fusion network, capable of obtaining more accurate cloud-removed images.
2) We propose the FAT network, which can generate more accurate translated images by mitigating interference on common features.
3) We propose the CGF network, which can treat features in cloudy and non-cloud regions differently, enabling further refinement of restored information in the translated images.

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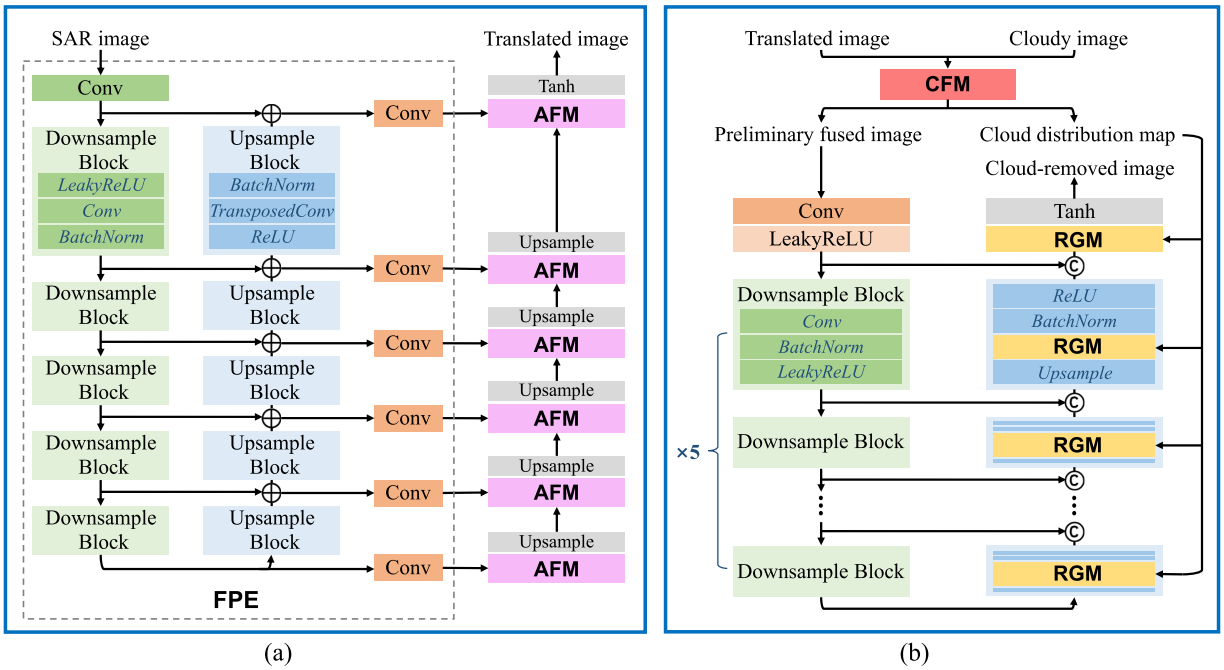


Fig. 2. Overview of the proposed SAR-assisted TCR method, which consists of (a) FAT network and (b) CGF network.

II. Related Works

A. Multitemporal TCR

Cloud cover changes over time, leading to varied cloudy regions across different periods. Multitemporal methods utilize images acquired at other times as reference data, leveraging the correspondence between them to restore missing information. These methods can be roughly classified into four categories: cloning methods [3], [4], low-rank representation methods [5], [6], dictionary learning methods [7], [8], and deep learning methods [9], [10]. However, the applicability of multitemporal methods is constrained by the requirement for a short time interval between images.

B. TCR Based on One-Step Fusion

One-step fusion treats TCR as an image fusion task. And existing methods primarily concentrate on fusion strategies. For example, SAR-Opt-cGAN [26], DSen2-CR [17], and G-FAN [20] first concatenate SAR images with cloudy images. The concatenated images are then inputted into a network to predict the cloud-removed images. The network architecture in SAR-Opt-cGAN is based on UNet [39], in DSen2-CR is extended from EDSR [37], and in G-FAN [20] utilizes a graph aggregation network. TF-CRNet [12] extracts features from SAR images and cloudy images independently. Then the features are concatenated and fed into the subsequent network. GLF-CR [16] uses two parallel branches for modality-specific learning and constantly integrates features from each modality using transformer layers. As discussed in the introduction, one-step fusion requires internal learning of both cloud segmentation and SAR-to-optical translation, thereby raising the complexity of training.

C. TCR Based on Two-Step Fusion

Two-step fusion learns the SAR-to-optical translation and image fusion separately. It first translates SAR images into optical equivalents. Then the translated optical images are fused with cloudy images to generate cloud-removed images. Simulation-Fusion-GAN [1] utilizes a generative adversarial network (GAN) [27] for SAR-to-optical translation. The translated images and cloudy images are concatenated and inputted into another GAN to predict the cloud-removed images. Both GANs have the same architecture based on UNet [39]. DRIBs-GAN [2] similarly employs two identical GANs, incorporating dilated residual inception blocks (DRIBs) to expand the receptive field of UNet. However, Simulation-Fusion-GAN and DRIBs-GAN exhibit limited capabilities for the two challenges outlined in the introduction. For the first challenge, they employ addition or concatenation to fuse features from different scales, disregarding the differences between these features. This may disrupt the distribution of common features, leading to the weakening of common features after feature fusion. For the second problem, the used convolution operations treat features from different regions equally, hindering the full utilization of non-cloudy regions to refine the restored regions.

III. Methods

This section provides an overview of the new SAR-assisted TCR method, followed by introductions to the internal FAT, internal CGF, and the loss functions.

A. Overview

The new SAR-assisted TCR method is illustrated in Fig. 2. The FAT translates SAR images into corresponding optical

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images. Then the CGF fuses the translated images with cloudy images to predict cloud-removed images. The FAT and the CGF mitigate the first and second challenges in Section I, respectively.

B. FAT Network

The FAT is designed to leverage common features shared between SAR and optical images to improve SAR-to-optical translation. The FAT is mainly composed of an FPE and an AFM [as shown in Fig. 2(a)]. The FPE extracts effective multiscale features from SAR images. Because SAR images contain objects of multiple sizes, such that the representation ability of single-scale features is limited. The AFM aims to align the position distribution of common features before and after feature fusion, thereby mitigating the weakening of common features.

1) Feature Pyramid Encoder: The FPE is illustrated in the dashed box in Fig. 2(a). In current SAR-to-optical translation methods [40], [41], multiscale features are derived from intermediate features of the bottom-up pathway [illustrated by the green boxes in Fig. 2(a)]. We aim to improve the representation ability of multiscale features. Inspired by the feature pyramid network [23], initially developed for object detection, FPE integrates a top-down pathway [depicted by the blue boxes in Fig. 2(a)] and connects it laterally with the existing bottom-up pathway. Thus, the fusion of high-level features with low-level features considers both spatial and semantic information. Finally, a set of 1×1 convolutions is added [illustrated by the orange boxes in Fig. 2(a)] to further enhance the mapping capability.

The feature pyramid network employs nearest neighbor interpolation in the top-down pathway to generate higher resolution features. Since semantically stronger features are spatially coarser, nearest neighbor interpolation has limited capability to recover spatial high-frequency information. Therefore, FPE replaces the nearest neighbor interpolation with transposed convolution to compensate for the lack of spatial information during upsampling. The work [42] has proven the effectiveness of transposed convolutions for tasks such as image generation.

2) Alignment Fusion Module: The FAT uses a sequence of AFMs to fuse multiscale SAR features, converting them into optical features, and thereby generating the corresponding optical translated images. In a single AFM [as shown in Fig. 3(a)], it receives the fused feature F_{i-1} from the preceding layer. F_{i-1} is first normalized by a *BatchNorm* and then fused with the SAR feature F_{S_i} at this layer via an alignment layer (AF). The following *ReLU* and *Convolution* layer map the fused feature into the intermediate feature F_i of this layer. Due to potential ambiguity introduced by differences between features at different layers, the AL is employed to align the common features post-fusion.

The AL is illustrated in Fig. 3(b). Specifically, AL needs to identify the positions of common features within the feature map. It can be implemented by assigning weight to each position $\omega \in \mathbb{R}^{h\times w\times c}$, i.e., giving high weights to the positions that may be the common features, and giving medium or low

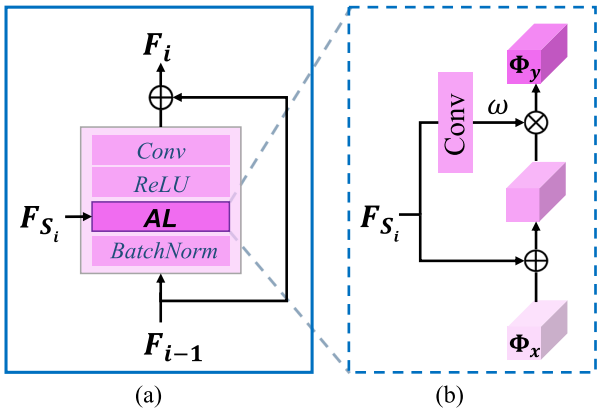


Fig. 3. Details of (a) AFM. F_{S_i} denotes the SAR feature. F_{i-1} is the feature from the preceding layer. F_i is the fused feature in this layer. ϕ_x and ϕ_y denote the input and output features of (b) AL, respectively.

weights to the rest. It is not easy to determine ω artificially so that ω is learnable. Since ω has the same dimension as F_{i-1} and F_{S_i}, ω represents the distribution of common features in the feature map. After that, AL adds F_{S_i} and F_{i-1} and multiplies ω with the added feature. The addition is to keep the feature dimension unchanged, and the multiplication is to ensure the distribution consistency of common features. The process of AL can be expressed as

$$\phi_y = \left(\phi_x + F_{S_i}\right) \odot \omega \tag{1}$$

where $\phi_x = \text{BatchNorm}(F_{i-1})$ and $\omega = \text{Conv}(F_{S_i})$.

C. CGF Network

The CGF distinguishes between cloudy and non-cloudy regions, utilizing information from non-cloudy regions in cloudy images to optimize the recovered information in translated images. As shown in Fig. 2(b), the CGF comprises a CFM and an RGM. The CFM predicts the cloud distribution map and generates the preliminary fused image. It considers the objective of cloud removal, preserving non-cloudy regions in cloudy images, while utilizing auxiliary images for recovering occluded regions. The RGM utilizes the cloud distribution map as spatial guidance and leverages information from non-cloudy regions to enhance the recovered regions.

1) Cloud-Aware Fusion Module: As shown in Fig. 4(a), given the cloudy image $\mathbf{C} \in \mathbb{R}^{h\times w\times c}$, the CFM predicts the cloud distribution map $\mathbf{M} \in \mathbb{R}^{h\times w\times 1}$ by a lightweight network (the lightweight network is illustrated in Fig. 4). The pixel values in M correspond to the probability of locations belonging to cloudy regions. Then, the CFM weights the cloudy images and the translated images T spatially according to M, which can be formulated as

$$F' = M \odot T + (E - M) \odot C \tag{2}$$

where $\mathbf{E} \in \mathbb{R}^{h\times w\times c_1}$ denotes the constant matrix with all the values as 1. Compared with the commonly used *concatenation*, the CFM has two advantages: 1) it introduces less irrelevant information in T and 2) it estimates M, which can serve as the spatial guidance for subsequent networks.

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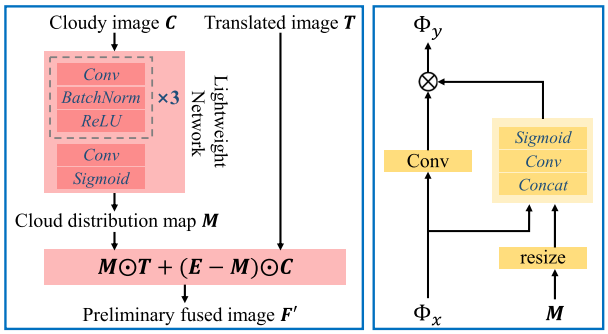


Fig. 4. Details of (a) CFM and (b) RGM. ϕ_x and ϕ_y denote the input and output features, respectively.

2) Region Gated Module: Current TCR networks utilize standard convolutions, which treat features from different locations as the same valid ones, ignoring the differences between cloudy and non-cloudy regions. To this end, the RGM aims to process features dynamically according to their associated regions. Gated convolution (GConv) [28] enables differential treatment of features through a gating mechanism. However, the network learns which features require disparate treatment, leaving this determination unconstrained. In the context of TCR, the RGM should prioritize features distinguishing between cloudy and non-cloudy regions. Therefore, based on GConv, the RGM uses M as prior information to dynamically process features in designated regions, which can be expressed as

$$\phi_y = \mathrm{Conv}(\phi_x) \odot \sigma(\mathrm{Conv}(\mathrm{Cat}(\phi_x, \mathrm{Down}(M)))) \tag{3}$$

where Cat denotes channel-wise concatenation, Down is bilinear interpolation downsampling, and $\sigma(\mathrm{Conv}(\mathrm{Cat}(\phi_x, \mathrm{Down}(M))))$ represents the gating mask.

D. Loss Functions

The loss functions used in FAT and CGF are basically the same, which consist of two items: the mean absolute error (MAE) loss $\mathcal{L}_{\mathrm{mae}}$ and the multiscale adversarial loss $\mathcal{L}_{\mathrm{madv}}$ [24].

1) MAE Loss: $\mathcal{L}_{\mathrm{mae}}$ measures the pixel-wise error to force the predictions to be near the ground truth in an L1 sense. Given a predicted image $\mathbf{X} \in \mathbb{R}^{h\times w\times c}$ and the ground truth $\mathbf{Y} \in \mathbb{R}^{h\times w\times c}$, $\mathcal{L}_{\mathrm{mae}}$ is defined as

$$\mathcal{L}_{\mathrm{mae}} = \frac{1}{N}\|\mathbf{X} - \mathbf{Y}\|_1 \tag{4}$$

where N denotes the total number of pixels.

2) Multiscale Adversarial Loss: Minimizing $\mathcal{L}_{\mathrm{mae}}$ alone can result in a reduction of high-frequency details and produce over-smoothed images. The adversarial loss $\mathcal{L}_{\mathrm{adv}}$ can capture perceptually relevant differences and generate images with high detail fidelity [22], which is achieved by adversarial training between the original network (namely G) and an additional discriminator network D. The discriminator is an independent network used to distinguish between real images and generated images. It can be combined with the original network to form adversarial training, thereby aiding the original network in generating richer textures and details. Adversarial training encourages the discriminator to accurately distinguish between real and generated images while also encouraging the original network to generate images that deceive the discriminator as much as possible. As the adversarial training progresses, the generated images become increasingly closer to real images. We use the same discriminator network as in [2]. Given a predicted image $\mathbf{X}$ and the ground truth $\mathbf{Y}$, $\mathcal{L}_{\mathrm{adv}}$ can be expressed as

$$\begin{aligned}\mathcal{L}_{\mathrm{adv}}(G, D) &= \mathbb{E}_{X,Y}\left[\log D(\mathbf{X}, \mathbf{Y})\right] + \\ &= \mathbb{E}_X\left[\log(1 - D(\mathbf{X}, G(\mathbf{X})))\right].\end{aligned} \tag{5}$$

To ensure global consistency, we extend $\mathcal{L}_{\mathrm{adv}}$ to multiscale adversarial loss $\mathcal{L}_{\mathrm{madv}}$, which utilizes two discriminators D_1 and D_2. Specifically, given $\mathbf{X}$ and $\mathbf{Y}$, D_1 works on the original scale while D_2 works on the images downsampled by a factor of 2. $\mathcal{L}_{\mathrm{madv}}$ can be formulated as

$$\mathcal{L}_{\mathrm{madv}}(G, D) = \sum_{k=1,2} \mathcal{L}_{\mathrm{adv}}(G, D_i). \tag{6}$$

In summary, the total loss function is written as

$$\mathcal{L}_{\mathrm{total}} = \lambda_1 \mathcal{L}_{\mathrm{mae}} + \lambda_2 \mathcal{L}_{\mathrm{madv}} \tag{7}$$

where λ_i is the weight of each loss item.

IV. Experiments

In this section, we describe the datasets, implementation details, and evaluation metrics. Since our method utilizes a two-step fusion framework, we conduct experiments on the FAT and the CGF separately to demonstrate the effectiveness of each network. We then compare our method with state-of-the-art methods to highlight its superior cloud removal performance.

A. Datasets

We conducted experiments on the SEN12MS-CR dataset [17], which is collected from real-world scenarios. To further validate the methods' effectiveness, we constructed a simulated dataset SIM-CR and performed the same experiments on it.

1) SEN12MS-CR Dataset: The SEN12MS-CR [17] is a multisource cloud removal dataset which is built upon the publicly released SEN12MS dataset [31]. It contains samples from most inhabited continents across all meteorological seasons. Each sample is composed of a Sentinel-1 SAR image with two polarization channels (VV and VH), a Sentinel-2 cloudy optical image with 13 original bands, and a Sentinel-2 cloud-free optical image. The cloudy and cloud-free images have values in the range [0, 10 000]. The ranges of the VV and VH channels in SAR images are $[-25, 0]$ and $[-32.5, 0]$, respectively. The image size is 256×256 pixels. We train and test the algorithms with 10 148 and 1118 samples, respectively.

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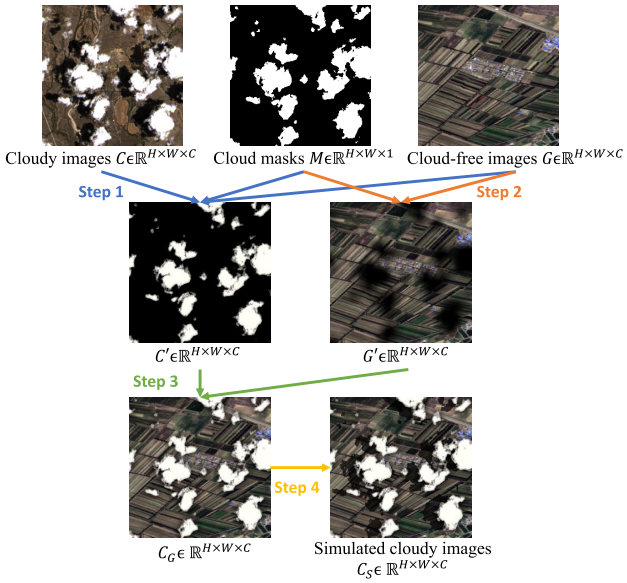


Fig. 5. Example of the simulation process for the SIM-CR dataset.

2) SIM-CR Dataset: Inspired by the work [2], the simulation process is illustrated in Fig. 5. Given a set of cloud-free images $G \in \mathbb{R}^{H\times W\times C}$, a set of cloudy images $C \in \mathbb{R}^{H\times W\times C}$, and the corresponding cloud masks $M \in \mathbb{R}^{H\times W\times 1}$, the simulated cloudy images $C_S \in \mathbb{R}^{H\times W\times C}$ are generated via the following four steps.

In the first step, clouds are extracted from the cloudy images C using the cloud masks M, and their color is adjusted to match the ambient color of the cloud-free images G. This process can be expressed by the following equation:

$$\begin{aligned} C'_{\text{gray}} &= M \times \text{GrayScale}(C) \\ [b_1, b_2, \ldots, b_c] &= [m_1, m_2, \ldots, m_c]/m \\ C' &= [b_1, b_2, \ldots, b_c] \cdot C'_{\text{gray}} \end{aligned} \tag{8}$$

where GrayScale denotes the operation of converting to grayscale, which reduces the influence of ambient color in the cloudy images C; m_i and m represent the mean of the ith band and the entire images in the cloud-free images G, respectively; $[b_1, b_2, \ldots, b_c]$ represents the relative intensity of each band in the cloud-free images G, which is used to approximate the distribution of the ambient color in G.

In the second step, we remove the surface information from the cloud-free images G that would be obscured using the cloud masks M, simulating the characteristic of thick clouds completely blocking surface data. Additionally, to ensure a smooth transition between the thick clouds and surface data, we apply a blur to the cloud masks M. This process is described by the following equation:

$$\begin{aligned} M_{\text{blur}} &= \text{Gau} * M \\ G' &= (1 - M_{\text{blur}}) \times G \end{aligned} \tag{9}$$

where Gau represents the Gaussian blur kernel, and G' represents the cloud-free images after removing the surface information.

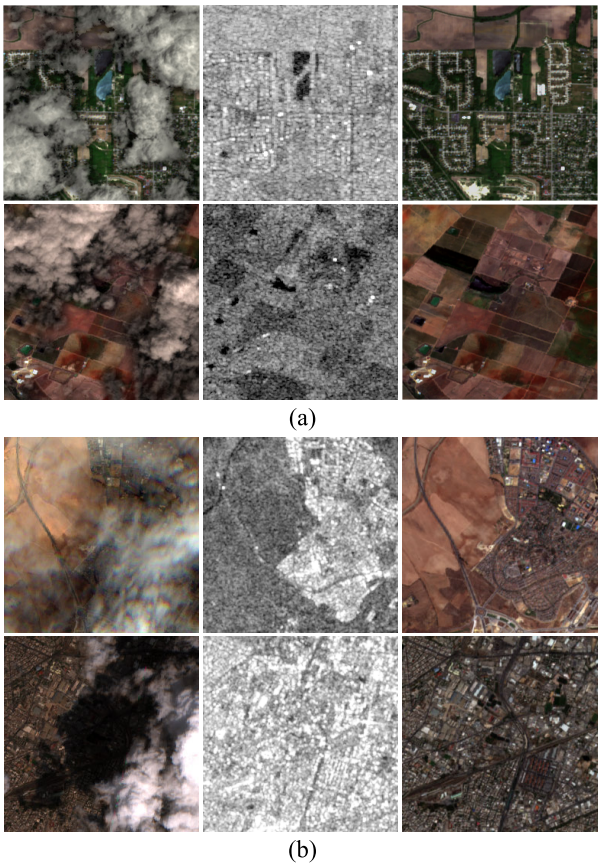


Fig. 6. Visualization samples of (a) SIM-CR and (b) SEN12MS-CR datasets. Each row represents a sample of triplet (cloudy image, SAR image, cloud-free image). The shown optical images are from the B4-B3-B2 bands. The shown SAR images are from the VV polarization channel.

In the third step, we incorporate the cloud information C' into G'. This process is represented by the following equation:

$$C_G = M_{\text{blur}} \times C' + G'. \tag{10}$$

In the fourth step, we simulate the shadows of thick clouds. Unlike thin clouds, the shadows of thick clouds typically mirror the shape of the clouds themselves and exhibit distinct edges. Hence, the cloud mask M is utilized to simulate these shadows. We randomly shift M to simulate shadow regions and multiply the spectral values of the shadow regions by the attenuation coefficient λ_{shadow}. After that, we derive the final simulated cloudy images C_S.

For the SIM-CR dataset, the cloudy images C and cloud masks M used in the simulation are taken from the 95-Cloud dataset [32], while the cloud-free images G are taken from the SEN1-2 dataset [33]. The λ_{shadow} is set to 0.3. Since the cloud-free images in the SEN1-2 dataset are paired with SAR images, the SIM-CR dataset can be used to validate SAR-assisted TCR methods. Both cloudy and cloud-free images contain RGB channels, while SAR images contain the VV polarization channel. All images maintain sizes of 256 × 256 pixels, with pixel values ranging from 0 to 255. We train and test the methods with 14 509 and 1623 samples, respectively. Fig. 6(a) and (b) show some examples of the SIM-CR and SEN12MS-CR datasets, respectively.

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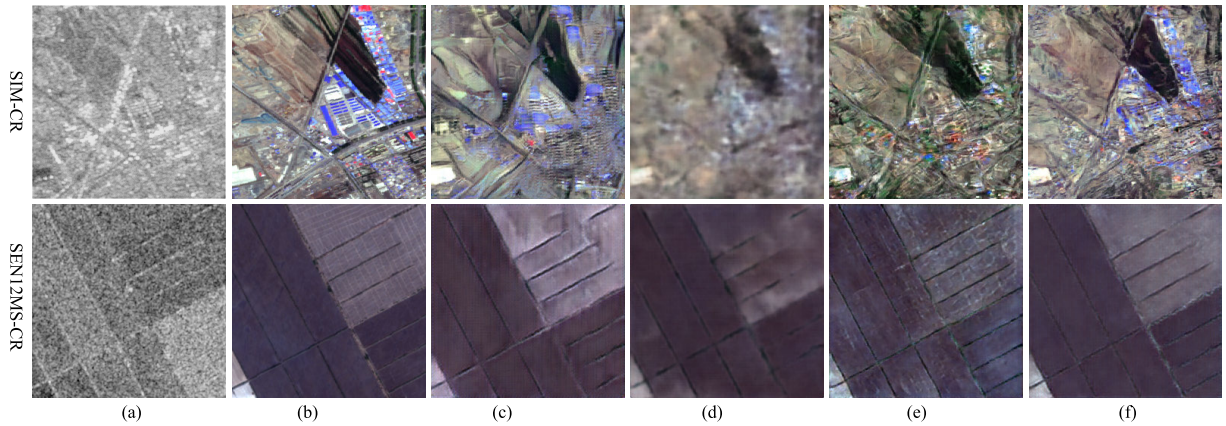


Fig. 7. Visualization of the translated images on the SIM-CR and SEN12MS-CR datasets (*zoom in for a better view*). The shown images are enhanced true-color RGB images. (a) SAR image. (b) Cloud-free image. (c) Translated image of SAR2OPT-CGAN [38]. (d) Translated image of the translation network in Simulation-Fusion-GAN [1]. (e) Translated image of the translation network in DRIBs-GAN [2]. (f) Translated image of our FAT.

B. Implementation Details

The proposed methods are implemented with PyTorch 1.12 on Python 3.8. All the experiments are trained with NVIDIA GeForce RTX 2080Ti GPUs. The size of the training images is 256×256 and the pixel values are normalized to $[-1, 1]$. We use the Adam optimizer and the batch size is set to 16. The networks are initialized with Glorot initialization [35]. We train the FAT with a two time-scale update rule (TTUR) [36] and the initial learning rate is 2×10^{-4}. The learning rate is linearly reduced to 0 after 100 epochs. The CGF is trained with a learning rate of 2×10^{-4} for 50 epochs. The momentum parameters in the Adam optimizer are $\beta_1 = 0.5$ and $\beta_2 = 0.999$. During experiments, we observed that setting different loss weights for the FAT and CGF, along with different implementation to the $\mathcal{L}_{\text{madv}}$, can lead to performance improvement. Specifically, for the FAT, the λ_1 and λ_2 in 7 are 100 and 1, respectively, and the $\mathcal{L}_{\text{madv}}$ is implemented with GAN hinge loss. For the CGF, the λ_1, λ_2, along with the implementation, are 50, 1, and GAN least square loss, respectively. Similar to [2], the FAT is trained first and then frozen before training the CGF.

The comparative experiments cover most of the open-source SAR-assisted TCR methods, including SAR2OPT-CGAN [38], SpA GAN [43], SAR-Opt-cGAN [26], DSen2-CR [17], and GLF-CR [16]. These methods were implemented with the source codes provided by their authors. We also reimplemented other methods according to their papers, including Simulation-Fusion-GAN [1], DRIBs-GAN [2], and TFCRNet [12]. The hyperparameters and training processes of all comparative methods are consistent with their original papers.

The proposed method requires 22.97 ms to predict a sample, with its model file occupying 156.17 MB of storage. Specifically, the FAT has an inference time of 17.70 ms and a storage cost of 99.07 MB, while the CGF's inference time and storage cost are 5.27 ms and 57.10 MB, respectively.

C. Metrics

We quantitatively evaluate the methods with five metrics: MAE, root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and spectral angle mapper (SAM). They are defined as

$$\text{MAE}(x, y) = \frac{1}{C \cdot H \cdot W} \sum_{c=h=w=1}^{C,H,W} \left| x_{c,h,w} - y_{c,h,w} \right| \tag{11}$$

$$\text{RMSE}(x, y) = \sqrt{\frac{1}{C \cdot H \cdot W} \sum_{c=h=w=1}^{C,H,W} \left(x_{c,h,w} - y_{c,h,w} \right)^2} \tag{12}$$

$$\text{PSNR}(x, y) = 20 \cdot \log_{10} \left(\frac{1}{\text{RMSE}(x, y)} \right) \tag{13}$$

$$\text{SSIM}(x, y) = \frac{\left(2\mu_x \mu_y + \epsilon_1\right)\left(2\sigma_{xy} + \epsilon_2\right)}{\left(\mu_x + \mu_y + \epsilon_1\right)\left(\sigma_x + \sigma_y + \epsilon_2\right)} \tag{14}$$

$$\text{SAM}(x, y) = \cos^{-1} \left(\frac{\sum_{c=h=w=1}^{C,H,W} x_{c,h,w} \cdot y_{c,h,w}}{\sqrt{\sum_{c=h=w=1}^{C,H,W} x_{c,h,w}^2 \cdot \sum_{c=h=w=1}^{C,H,W} y_{c,h,w}^2}} \right) \tag{15}$$

where $x, y \in \mathbb{R}^{C \times H \times W}$ are images to be compared, $x_{c,h,w}, y_{c,h,w}$ are rescaled to range $[0, 1]$, μ_x, μ_y denote mean values of images, σ_x, σ_y standard deviations, σ_{xy} covariance, and ϵ_1, ϵ_2 small numbers to stabilize the calculation. MAE and RMSE quantify the mean pixel-wise differences. PSNR reflects the degree to which a degraded image is reconstructed in the form of signal-to-noise ratio. SSIM measures the structural similarity that is independent of the average luminance and contrast. SAM quantifies the spectral similarity by calculating angles between spectral vectors.

D. Experimental Results on the Translation Network

In the two-step fusion framework, the translation network is responsible for SAR-to-optical translation. To verify the effectiveness of the FAT network, we exhibit the comparative results with other SAR-to-optical translation networks. Furthermore, to assess the impact of the internal FPE and AFM, we conducted experiments by removing or disabling them, which is referred to as ablation studies.

1) Comparative Results: We compare the FAT with SAR2OPT-CGAN [38], the translation network in

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TABLE I
COMPARATIVE RESULTS FOR THE TRANSLATION NETWORK ON THE SIM-CR AND SEN12MS-CR DATASETS

Datasets	**SIM-CR**					**SEN12MS-CR**				
Metrics	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
SAR2OPT-CGAN [38]	0.1127	0.1606	15.9742	0.3091	8.7843	0.0205	0.0327	30.0481	0.8664	4.992
Sim-Fus-GAN [1]	0.0914	0.1293	17.8620	0.3863	6.7640	0.0191	0.0307	30.5752	0.8733	4.7462
DRIBs-GAN [2]	0.0833	0.1222	18.3901	0.4591	7.4081	0.0177	0.0277	31.3921	0.8761	4.7713
FAT (Ours)	**0.0759**	**0.1088**	**19.2671**	**0.4635**	**6.3652**	**0.0165**	**0.0255**	**31.8553**	**0.8808**	**4.1921**

TABLE II
ABLATION RESULTS TOWARD THE FPE AND AFM ON THE SIM-CR AND SEN12MS-CR DATASET

Datasets		**SIM-CR**					**SEN12MS-CR**				
FPE	**AFM**	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
		0.0854	0.1213	18.3241	0.3928	6.8141	0.0181	0.0285	31.2234	0.8485	4.8112
✓		0.0819	0.1163	18.6862	0.4268	6.6362	0.0175	0.0279	31.4372	0.8659	4.6894
	✓	0.0779	0.1115	19.0570	0.4484	6.5100	0.0169	0.0268	31.7537	0.8748	4.5257
✓	✓	**0.0759**	**0.1088**	**19.2671**	**0.4635**	**6.3652**	**0.0165**	**0.0255**	**31.8553**	**0.8808**	**4.1921**

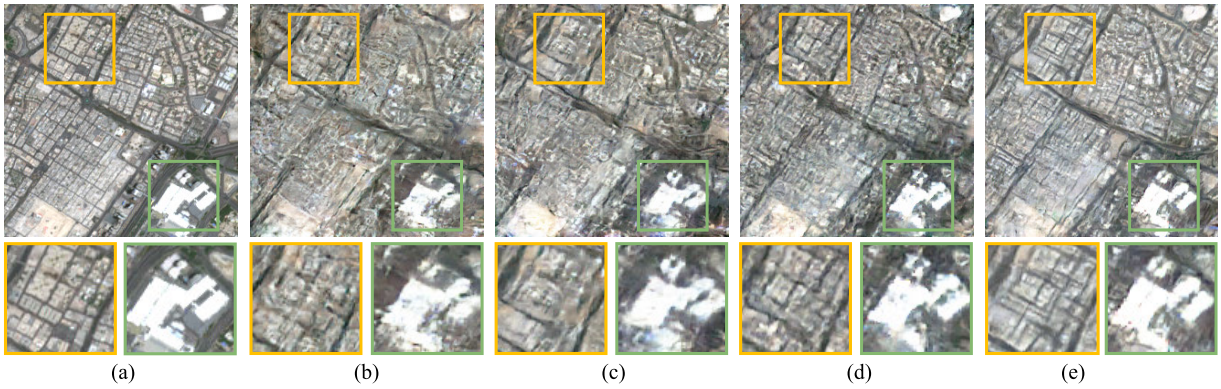


Fig. 8. Visualization of the translated images in the ablation studies toward the FPE and AFM. (a) Cloud-free image. (b) Baseline. (c) Baseline + FPE. (d) Baseline + AFM. (e) Baseline + FPE + AFM (FAT).

Simulation-Fusion-GAN [1], and the translation network in DRIBs-GAN [2]. SAR2OPT-CGAN models the translation with stacks of residual blocks. Simulation-Fusion-GAN utilizes UNet and learns only with L1 loss. DRIBs-GAN improves the UNet with DRIBs and uses SSIM in addition to the L1 loss. Table I shows the quantitative results and Fig. 7 visualizes the translated images.

SAR2OPT-CGAN fails to generate fine-grained details because it processes features with a single resolution. Although Simulation-Fusion-GAN introduces multiscale features, the results are over-smoothed because it only uses L1 loss, which is limited in its ability to capture high-texture details. DRIBs-GAN performs better as it uses adversarial loss and SSIM loss in addition to the L1 loss. Besides, its DRIBs increase the receptive field, helping the network to learn more patterns. However, as shown in Fig. 7(e), there are visual artifacts because it has a limited capacity to deal with the inconsistency between SAR features and optical features. Our FAT alleviates the degradation using FPE and AFM. The quantitative and qualitative results show that the FAT can get translated images with high fidelity.

2) Ablation Studies: We tested the experimental results by individually removing the FPE or AFM, and by simultaneously removing the FPE and AFM, to verify the effectiveness of each module in the FAT. For simplicity, this kind of experimental process is referred to as ablation testing in Section IV-E. Table II shows the quantitative results. The second row without FPE and AFM means the baseline UNet network. Referring to Fig. 2(a), the baseline network eliminated the upsample blocks in encoder and only retrain the downsample blocks. Additionally, it substituted vanilla convolutional layers for the AFM in the decoder. Fig. 8 shows the translated images. As revealed in the table, the AFM plays a key role in the FAT, contributing a ∼8.7% improvement. We replace the encoder in baseline with the FPE, and the results show that the FPE achieves a ∼4.1% improvement. After adding both the FPE and AFM, we achieve a performance improvement of ∼11.1% compared to the baseline. Based on the visualization results in Fig. 8, we can draw the following conclusions. The FPE improves the representation of the features and generates clearer results. Comparing Fig. 8(b) with (c), without the FPE (column b), the results are less meaningful, filled with subtle artifacts. In addition, the AFM exhibits its effectiveness in capturing the structural features (we think that structural features are one of the common features between SAR and optical images). As shown in Fig. 8(d), the outlines of roads

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TABLE III
COMPARATIVE RESULTS FOR THE FUSION NETWORK ON THE SIM-CR AND SEN12MS-CR DATSETS

Datasets	**SIM-CR**					**SEN12MS-CR**				
Metrics	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
Sim-Fus-GAN [1]	0.0408	0.0698	23.3014	0.7866	3.6294	–	–	–	–	–
DRIBs-GAN [2]	0.0405	0.0706	23.1883	0.7885	3.6542	0.0116	0.0189	34.6592	0.9344	3.3991
CGF (ours)	**0.0361**	**0.0622**	**24.1261**	**0.8038**	**3.3733**	**0.0106**	**0.0170**	**35.4070**	**0.9393**	**2.9641**

TABLE IV
ABLATION RESULTS TOWARD THE CFM AND RGM ON THE SIM-CR DATASET

Cat	**CFM**	**Conv**	**GConv**	**RGM**	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
✓		✓			0.0443	0.0695	23.1544	0.7777	4.0132
	✓	✓			0.0423	0.0648	23.7651	0.7741	3.8513
✓			✓		0.0395	0.0649	23.7514	0.7977	3.4731
	✓		✓		0.0379	0.0643	23.8413	0.7936	3.4712
	✓			✓	**0.0361**	**0.0622**	**24.1216**	**0.8038**	**3.3733**

TABLE V
ABLATION RESULTS TOWARD THE CFM AND RGM ON THE SEN12MS-CR DATASET

Cat	**CFM**	**Conv**	**GConv**	**RGM**	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
✓		✓			0.0125	0.0190	34.4080	0.9240	3.7992
	✓	✓			0.0122	0.0183	34.7521	0.9246	3.4775
✓			✓		0.0113	0.0178	35.2634	0.9313	3.2097
	✓		✓		0.0108	0.0173	35.4041	0.9358	2.9553
	✓			✓	**0.0106**	**0.0170**	**35.4070**	**0.9393**	**2.9641**

and buildings become sharper and more consistent with the cloud-free image.

E. Experimental Results on the Fusion Network

In the two-step fusion framework, the fusion network is responsible for fusing the translated images and the cloudy images to get the cloud-removed images. To verify the effectiveness of the CGF network, we compare the results against other fusion networks. We also conducted ablation studies toward the internal CFM and RGM.

1) Comparative Results: We compare our CGF with the fusion network in Simulation-Fusion-GAN [1] and the fusion network in DRIBs-GAN [2]. Both networks use the same architecture as their translation networks. For a fair comparison, *the translated images used in different methods are the same,* i.e., the translated images from our FAT. Table III shows the quantitative results, and Fig. 9 shows the cloud-removed images. The results of Simulation-Fusion-GAN on the SEN12MS-CR dataset are not presented because it uses another pretrained network to construct a perceptual loss function [25]. The network is pretrained on images with three bands (RGB) and cannot be used on images in SEN12MS-CR that have 13 bands.

For Simulation-Fusion-GAN and DRIBs-GAN, the recovered textures in cloudy regions are blurred. These methods fuse the cloudy and translated images with a global channel-wise concatenation. However, the information of the translated images in non-cloudy regions is actually redundant, the concatenation introduces this redundancy, causing subsequent networks to distinguish between valid and redundant information. Besides, they give less attention to the differences between cloudy and non-cloudy regions, which limits their restoration. Both quantitative and qualitative results demonstrate that our CGF can better recover the cloud-occluded information.

2) Ablation Studies: Ablation studies are conducted to verify the CFM and RGM. Tables IV and V show the quantitative results, in which *Cat* denotes *concatenation*. As the RGM is extended from GConv, we also add the ablation comparison against GConv. The RGM can only be equipped with the CFM because it needs the cloud distribution map from the CFM. As revealed in the table, the CFM performs better than Cat no matter whether Conv or GConv is used. And the best results are achieved using both the CFM and RGM. Fig. 10 visualizes the ablation results. As shown in the first row, without the CFM and RGM (column c), the network produces a blur effect in the recovered region (green boxes). In the second row, the baseline exhibits more errors in the non-cloudy region (magenta boxes). This phenomenon demonstrates that the CFM helps to introduce less redundant data, keeping the non-cloudy region intact to a large extent. Moreover, thanks to the RGM, more textures are reconstructed.

3) Visualization of the Cloud Distribution Map: The CFM realizes the localization of cloudy regions to guide the image fusion. To verify that the CFM learns the correct cloudy regions, Fig. 11 visualizes the learned cloud distribution map M. It can be seen that M successfully fits the distribution of cloudy areas, even though the corresponding labels (i.e., cloud

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Fig. 9. Visualization results for the fusion network on the SIM-CR and SEN12MS-CR datasets (*zoom in for a better view*). The shown images are enhanced true-color RGB images. (a) Cloudy image. (b) Translated image. (c) Cloud-free image. (d) Cloud-removed image of the fusion network in Simulation-Fusion-GAN [1]. (e) Cloud-removed image of the fusion network in DRIBs-GAN [2]. (f) Cloud-removed image of our CGF network. *The translated image in different methods are the same. The result of Simulation-Fusion-GAN on SEN12MS-CR is not presented because its perceptual loss cannot be applied on images with 13 bands.*

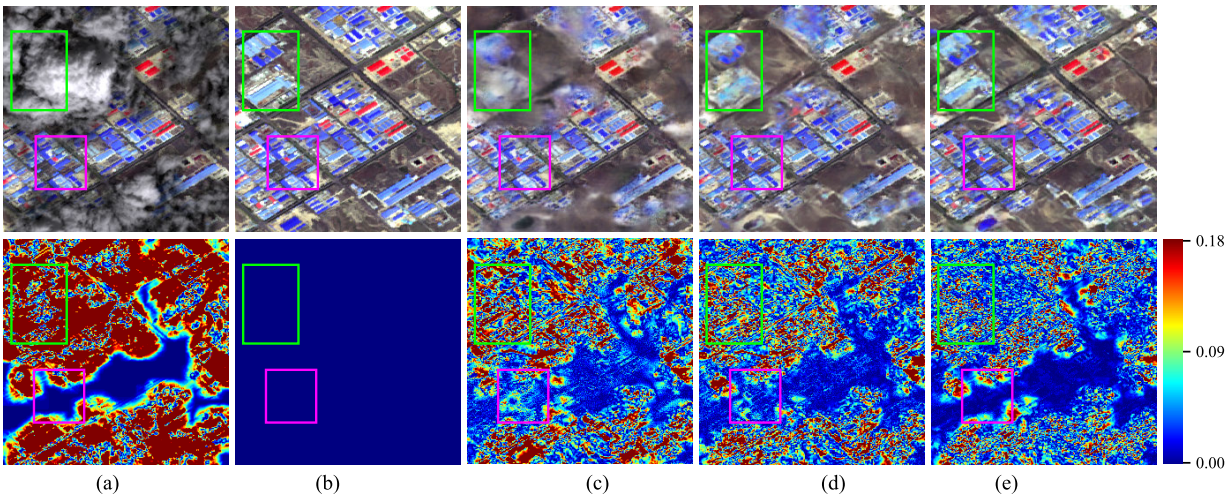


Fig. 10. Visualization results in the ablation studies toward the CFM and RGM. The first row represents the cloud-removed images. The second row visualizes the differences between images in the first row and the cloud-free images. (a) Cloudy image. (b) Cloud-free image. (c) Cloud-removed image of UNet baseline. (d) Cloud-removed image of baseline with the CFM. (e) Cloud-removed image of baseline with the CFM and RGM (CGF). To enhance visualization, errors greater than 0.18 are still displayed using the color corresponding to 0.18.

masks) are not provided during training. This demonstrates the rationality of CFM modeling.

F. Experimental Results on TCR

After verifying the FAT and CGF independently, we compare our TCR method with other methods to show our advantages. The compared methods include the SAR-to-optical translation approach: SAR2OPT-CGAN [38]; the multispectral-based approaches: SpA GAN [43]; the one-step fusion approaches: SAR-Opt-cGAN [26], TF-CRNet [12], DSen2-CR [17], GLF-CR [16]; and the two-step fusion approaches: Simulation-Fusion-GAN [1], DRIBs-GAN [2]. Table VI shows the quantitative results. Figs. 12 and 13 visualize the cloud-removed images. The results of Simulation-Fusion-GAN on the SEN12MS-CR dataset are not presented because it uses another pretrained network to construct a perceptual loss function. The network is pretrained on images with three bands (RGB) and cannot be used on images in the SEN12MS-CR dataset that have 13 bands.

According to Table VI, our method achieves the best results. First, we can find that methods based on SAR-optical fusion outperform those using only SAR (SAR2OPT-CGAN) or only cloudy images (SpA GAN). The possible reasons are as follows. SAR images primarily capture the structural properties of the surface. Therefore, SAR2OPT-CGAN, which relies solely on SAR images, can recover the main geometric information but struggles to restore the local color variations. SpA GAN, which uses only cloudy images, can only recover visually plausible but fake information. Because the thick clouds totally block the surface and the blocked information is lost. In contrast, methods based on SAR-optical fusion can utilize SAR to provide occluded information and use the cloudy images to offer rich colors and textures, thus achieving better results. Then, based on the visualization results in Figs. 12 and 13, we can see that the two-step fusion methods (i.e., Sim-Fus-GAN, DRIBs-GAN, and Ours) recover more details than the one-step fusion methods (i.e., SAR-Opt-cGAN, DSen2-CR, TF-CRNet, and GLF-CR). We explain the reason as: the

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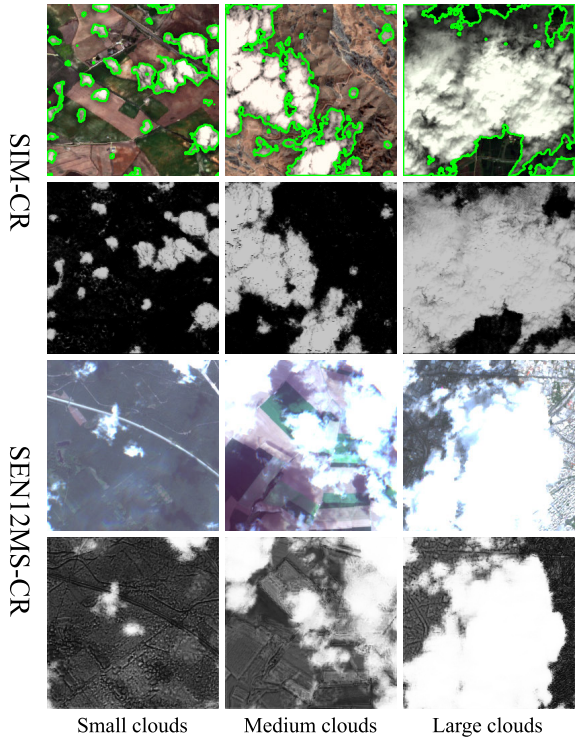


Fig. 11. Visualization of the learned cloud distribution map. The first and third rows show the cloudy images from the SIM-CR and SEN12MS-CR datasets, respectively. The second and fourth rows show the predicted cloud distribution maps. In the SIM-CR dataset, synthetic cloud boundaries are highlighted in green.

single fusion network learns three tasks simultaneously including cloud segmentation, SAR-to-optical translation and global consistency maintenance. Due to the different characteristics of tasks, it is difficult to ensure that all the tasks obtain the best performance. For example, if the SAR-to-optical translation is not learned well internally, the restored texture and details will be compromised. On the contrary, the two-step algorithm learns the SAR-to-optical translation separately, fully utilizing the information of SAR. However, they still face the problems like the heterogeneous characteristics between SAR and optical features and the unbalanced treatment between cloudy and non-cloudy areas. Our method relieves the problems and achieves higher performance.

G. Results Between Training Independently and End-to-End

As described in Section IV-B, we trained the FAT independently, then freezing its parameters and training the CGF. However, the FAT and CGF can also be trained end-to-end, which means the FAT is not frozen and it is trained with the CGF as a whole model. To compare these two training strategies, Table VII presents the comparative results. It can be seen that on the SIM-CR and SEN12MS-CR datasets, the metrics of independently training generally outperform the end-to-end training.

H. Results of Validating the Effectiveness of the Simulation

In Section IV-A, we designed a data simulation process and constructed the SIM-CR dataset to further verify the effectiveness of various methods. To explore the application value of this simulation process and dataset, we conducted experiments considering practical application scenarios. Existing methods depend on a large number of paired cloudy and cloud-free images for network training. However, collecting these paired images is costly in practical scenarios. Thus, it is beneficial to use simulated images to augment the limited real images for training. To this end, we designed two experiments.

For the first experiment, we randomly selected 100 real training samples from the SEN12MS-CR dataset to simulate scenarios where the cost of collecting real samples is high. Subsequently, we randomly selected 3000 simulated samples from the SIM-CR dataset used as supplementary training samples. Since the SEN12MS-CR and SIM-CR datasets differ in the number of bands and value ranges (specifically, the number of channels and value ranges for optical images in the SEN12MS-CR dataset are 13 and [0, 10 000], respectively; while for SAR, they are 2, [−25, 0], and [−32.5, 0], respectively. In the SIM-CR dataset, the number of channels for optical images and SAR images are 3 and 1, respectively, and the value ranges are both [0, 255]), for the optical images and SAR images in the SEN12MS-CR dataset, we, respectively, took the RGB and VV channels and scaled the value ranges to [0, 255]. Table VIII presents the cloud removal results of the proposed TCR method on the SEN12MS-CR testing set before and after adding simulated training samples. It can be observed that after adding simulated samples, each indicator improves by 6.88%, 4.85%, 2.62%, 3.22%, and 3.20%, respectively. This verifies the effectiveness of the SIM-CR dataset.

For the second experiment, considering the different number of bands and value ranges between the SIM-CR and SEN12MS-CR datasets, we used the original 100 real samples to generate the additional 3000 simulated samples. We obtained cloud masks for the 100 real samples using a cloud detector from [44]. For each cloud-free image in the real samples, we randomly selected 30 cloud masks and their corresponding clouds, generating 30 additional simulated cloudy images following the simulation process outlined in Section IV-A. Table IX presents the cloud removal results of the proposed TCR method on the SEN12MS-CR testing set before and after adding simulated training samples. It can be observed that after adding simulated samples, each indicator improves by 41.70%, 33.96%, 15.27%, 4.05%, and 24.26%, respectively. This verifies the effectiveness of the simulation process.

V. Discussion

A. How Cloud Information Benefits TCR

The intuitive principle of SAR-assisted TCR is to recover the occluded optical data with SAR data and keep the unoccluded data. Therefore, the spatial information of clouds is very important. Many previous networks [2], [16], [17], [26] focus on the fusion strategies, ignoring the modeling of cloudy regions. In the CGF, both the CFM and RGM use cloud information, i.e., the cloud distribution map. According to the ablation studies in Table IV and Fig. 10, the cloud distribution map can help the network to differentiate the features between cloudy and non-cloudy regions thus enabling local restoration.

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TABLE VI
COMPARATIVE RESULTS FOR TCR ON THE SIM-CR AND SEN12MS-CR DATASETS

Datasets	**SIM-CR**					**SEN12MS-CR**				
Metrics	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
SAR2OPT-CGAN [38]	0.1127	0.1606	15.9742	0.3091	8.7843	0.0205	0.0327	30.0481	0.8664	4.9921
SpA GAN [43]	0.0819	0.1296	18.7784	0.6247	6.0438	0.0325	0.0475	26.8219	0.8367	8.5894
SAR-Opt-cGAN [26]	0.0641	0.1086	19.4593	0.6620	4.9392	0.0167	0.0265	31.7512	0.8868	4.4092
DSen2-CR [17]	0.0377	0.0779	22.1734	0.7892	**2.7793**	0.0134	0.0211	33.5184	0.9288	3.3361
TF-CRNet [12]	0.0436	0.0862	21.4391	0.7595	3.3092	0.0133	0.0220	33.4892	0.9289	3.2653
GLF-CR [16]	0.0415	0.0636	23.9372	0.7535	2.8551	0.0111	0.0175	35.1350	0.9313	2.9982
Sim-Fus-GAN [1]	0.0460	0.0789	22.2021	0.7468	3.6350	-	-	-	-	-
DRIBs-GAN [2]	0.0419	0.0748	22.7044	0.7831	3.6662	0.0116	0.0189	34.6592	0.9344	3.3992
Ours	**0.0361**	**0.0622**	**24.1261**	**0.8038**	3.3733	**0.0106**	**0.0170**	**35.4070**	**0.9393**	**2.9641**

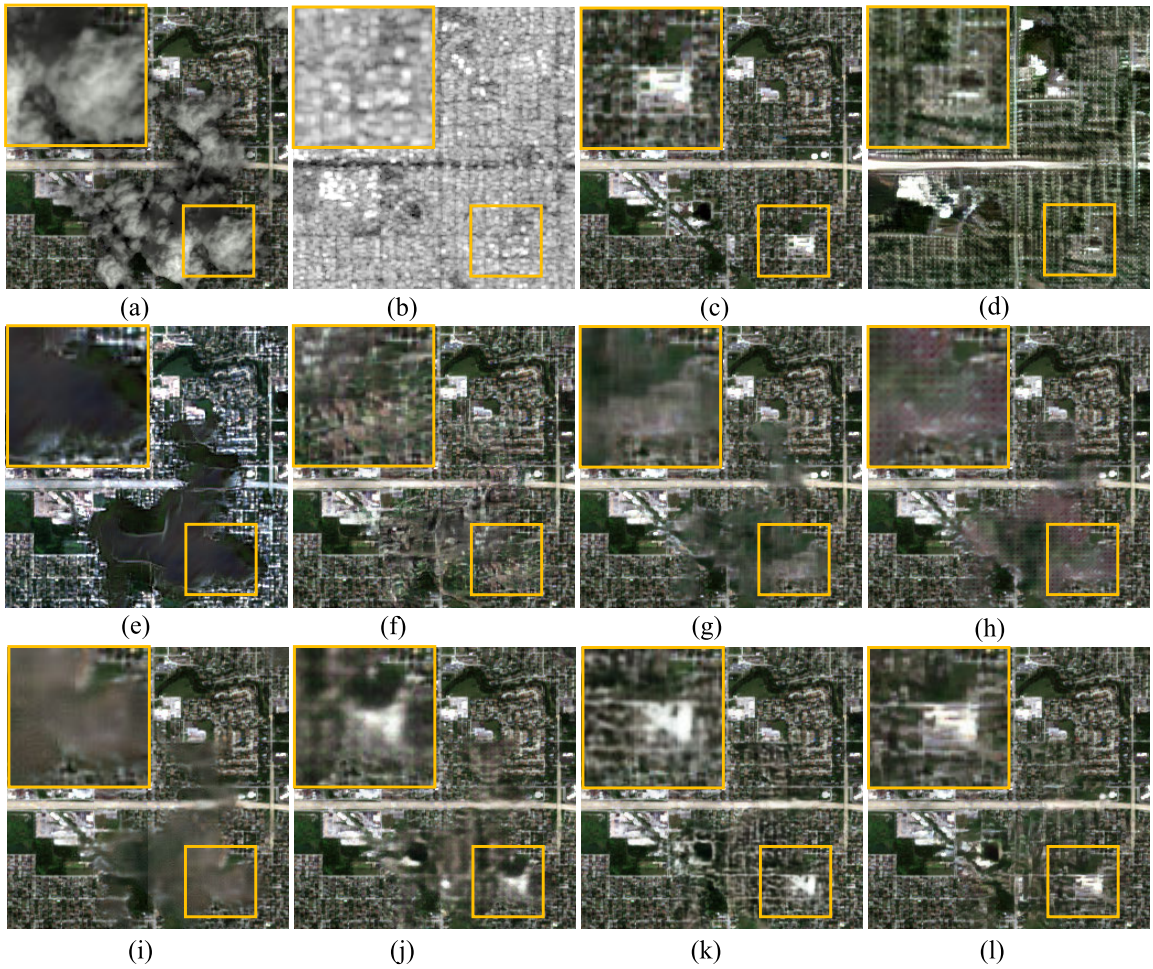


Fig. 12. Visualization of the cloud-removed images on the SIM-CR dataset (*zoom in for a better view*). The shown images are enhanced true-color RGB images. (a) Cloudy image. (b) SAR image. (c) Cloud-free image. (d) SAR2OPT-CGAN [38]. (e) SpA GAN [43]. (f) SAR-Opt-cGAN [26]. (g) DSen2-CR [17]. (h) TF-CRNet [12]. (i) GLF-CR [16]. (j) Simulation-Fusion-GAN [1]. (k) DRIBs-GAN [2]. (l) Ours.

B. Why Training Independently Works Better

According to the results in Table VII, we can find that training the FAT and CGF independently works better than training them end-to-end. A likely explanation is that, during the initial stage of end-to-end training, the FAT encounters challenges in producing stable translated images, thereby impeding the training of CGF. The CGF learns to fuse translated images with cloudy images to generate cloud-removed images. In the early stage of end-to-end training, the translated images generated by the FAT are not stable enough. These irregular translated images make it difficult for the CGF to grasp the relationship between cloudy images and translated images, thereby disrupting the training stability. In contrast, independent training ensures that the CGF always receives stable translated images, thus achieving better training results.

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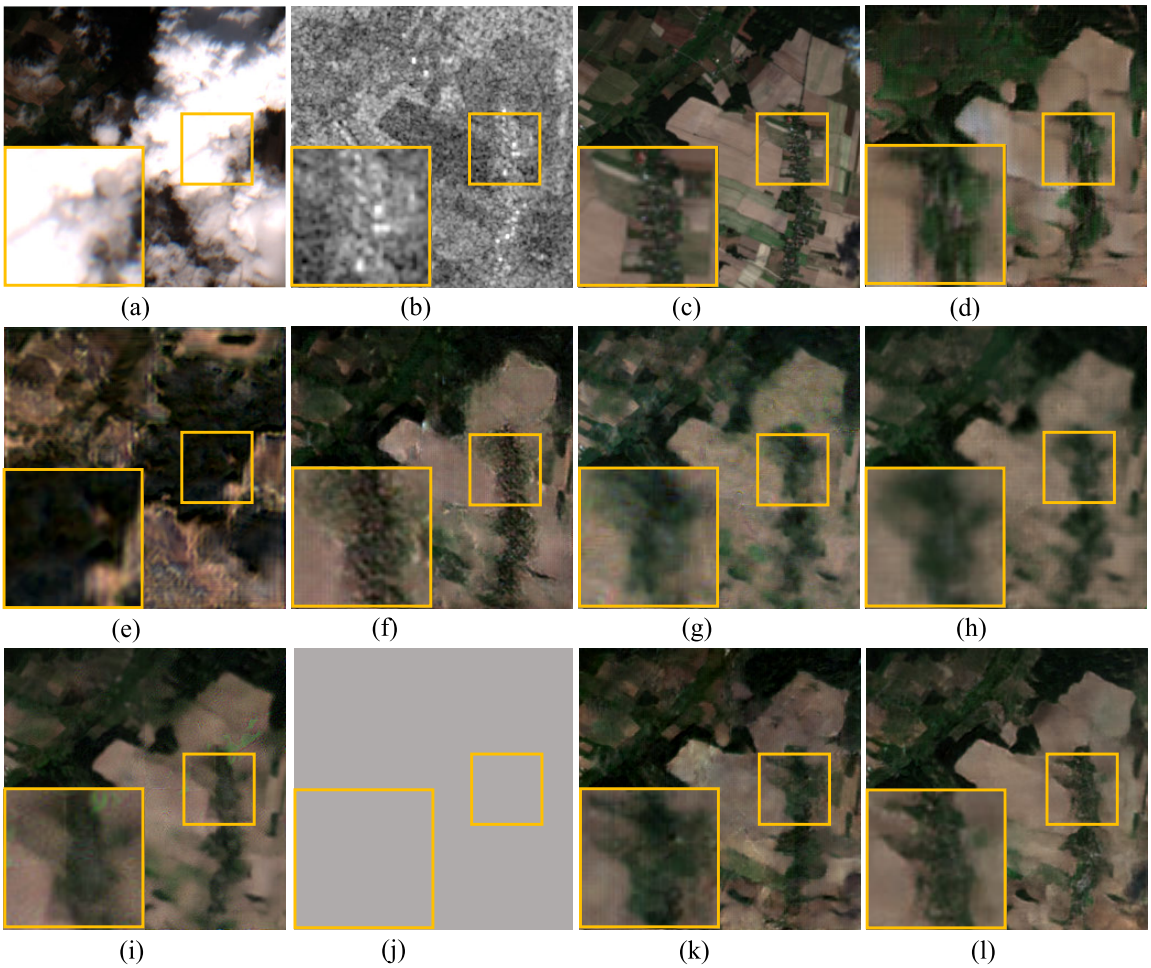


Fig. 13. Visualization of the cloud-removed images on the SEN12MS-CR dataset (*zoom in for a better view*). The shown images are enhanced true-color RGB images. (a) Cloudy image. (b) SAR image. (c) Cloud-free image. (d) SAR2OPT-CGAN [38]. (e) SpA GAN [43]. (f) SAR-Opt-cGAN [26]. (g) DSen2-CR [17]. (h) TF-CRNet [12]. (i) GLF-CR [16]. (j) Simulation-Fusion-GAN [1]. (k) DRIBs-GAN [2]. (l) Ours. *The results of Simulation-Fusion-GAN are not presented because its perceptual loss cannot be applied to the SEN12MS-CR dataset whose images have 13 bands.*

TABLE VII

COMPARATIVE RESULTS BETWEEN TRAINING INDEPENDENTLY AND END-TO-END

Datasets	**SIM-CR**					**SEN12MS-CR**				
Metrics	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
End-to-end	0.0375	0.0701	23.2798	0.7805	3.3399	0.0110	0.0173	35.3824	0.9337	2.9992
Independently	**0.0361**	**0.0622**	**24.1261**	**0.8038**	**3.3733**	**0.0106**	**0.0170**	**35.4070**	**0.9393**	**2.9641**

TABLE VIII

RESULTS OF VALIDATING THE EFFECTIVENESS OF THE SIM-CR DATASET. × MEANS ONLY USING 100 REAL SAMPLES FOR TRAINING, WHILE ✓MEANS ADDING THE SIMULATED SAMPLES

	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
×	0.1018	0.1424	17.1151	0.5038	3.3832
✓	**0.0948**	**0.1355**	**17.5634**	**0.5200**	**3.2751**

TABLE IX

RESULTS OF VALIDATING THE EFFECTIVENESS OF THE SIMULATION PROCESS. × MEANS ONLY USING 100 REAL SAMPLES FOR TRAINING, WHILE ✓MEANS ADDING THE SIMULATED SAMPLES

	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	SAM ↓
×	0.0470	0.0639	24.2472	0.8326	8.4933
✓	**0.0274**	**0.0422**	**27.9491**	**0.8663**	**6.4325**

C. Effectiveness of the Simulation Process

The simulated SIM-CR dataset is primarily used to validate methods across different datasets. The cloud-free images and SAR images used in the simulation process differ from those in the real SEN12MS-CR dataset. Nonetheless, Table VIII indicates that adding training samples from the SIM-CR dataset can improve performance to some extent. If we use data from the same source for simulation, the resulting simulated samples will significantly enhance the training effect. As shown in Table IX, after adding the simulated samples,

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the MAE indicator improved by 41.70% compared to before the addition. This validates the effectiveness of the simulation process.

D. Limitations and Improvement Directions

The two-step fusion framework has an increased number of parameters due to the use of two networks. Future research can consider how to address the problems in the one-step framework. Besides, we find that the explicit modeling of cloudy regions plays a positive role. How to integrate cloud detection and cloud removal together is another valuable research direction.

For the proposed SIM-CR dataset, the SAR and optical images are from Sentinel-1 and Sentinel-2 missions, respectively, which means models trained on this dataset may break down when used on other satellites. In addition, for simplicity, only vertically polarized (VV) band and three visible bands are considered in SAR and optical images, respectively, while the Sentinel-1 actually provides two SAR bands (i.e., VV and VH) and the Sentinel-2 provides 13 optical bands. The intention of SIM-CR is to help researchers to verify approaches comprehensively because SIM-CR does not contain the variations in ambient light, surface conditions, etc. Besides, it is more challenging as all the clouds are thick clouds, which means the information under the clouds is totally missing. Another advantage of SIM-CR is that the precise cloud masks are provided, which is necessary for some cloud removal methods.

VI. Conclusion

For SAR-assisted TCR, the two-step fusion is an effective framework. The large feature gap between SAR and optical images, and the underestimation of the spatial relationship between cloudy and non-cloudy regions, are two problems of the current two-step fusion methods. This article proposes a new SAR-assisted TCR algorithm based on the two-step fusion framework. In the FAT network, the FPE aims to enrich the capacities of multiscale SAR features. The AFM aims to align the common features before and after feature fusion. In CGF network, the CFM aims to estimate the spatial distribution M of cloudy regions and reduce the redundant information. The RGM aims to treat the features between cloudy and non-cloudy regions differently under the guidance of M. The ablation experiments and comparison experiments demonstrate the effectiveness of each module and network, respectively. Its limitations and improvement directions are also discussed.

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